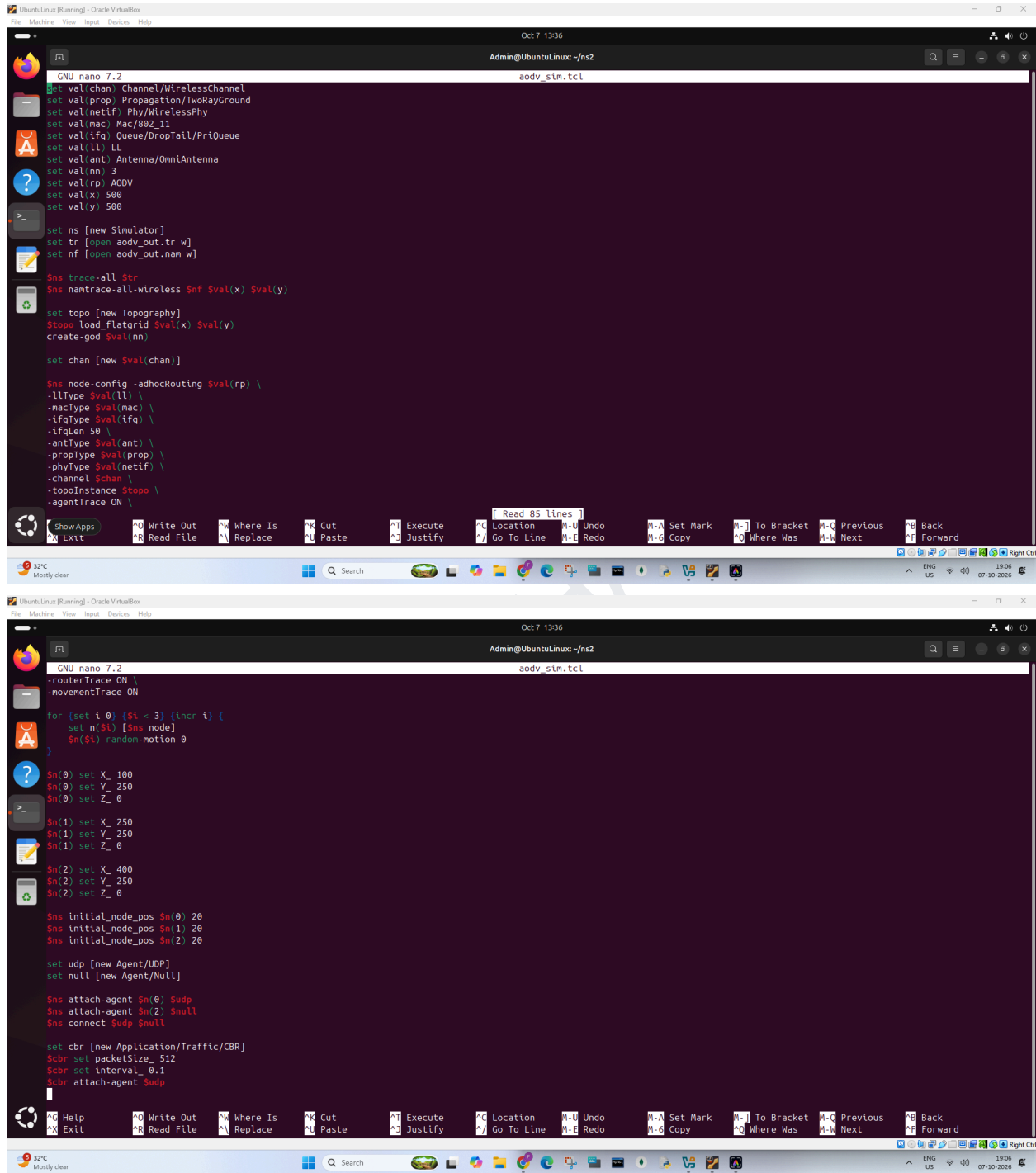


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AIM: AODV in NS2

Nano code:



```
GNU nano 7.2 aodv_sim.tcl
set val(chan) Channel/WirelessChannel
set val(prop) Propagation/TwoRayGround
set val(netif) Phy/WirelessPhy
set val(mac) Mac/802_11
set val(ifq) Queue/DropTail/PriQueue
set val(ll) LL
set val(ant) Antenna/OmnAntenna
set val(nn) 3
set val(rp) AODV
set val(x) 500
set val(y) 500

set ns [new Simulator]
set tr [open aodv_out.tr w]
set nf [open aodv_out.nam w]

$ns trace-all $tr
$ns namtrace-all-wireless $nf $val(x) $val(y)

set topo [new Topography]
$topo load_flatgrid $val(x) $val(y)
create-god $val(nn)

set chan [new $val(chan)]

$ns node-config -adhocRouting $val(rp) \
  -llType $val(ll) \
  -macType $val(mac) \
  -ifqType $val(ifq) \
  -ifqLen 50 \
  -antType $val(ant) \
  -propType $val(prop) \
  -phyType $val(netif) \
  -channel $chan \
  -topoInstance $topo \
  -agentTracesOn

for {set i 0} {$i < 3} {incr i} {
  set n($i) [$ns node]
  $n($i) random-motion 0
}

$ns(0) set X_ 100
$ns(0) set Y_ 250
$ns(0) set Z_ 0

$ns(1) set X_ 250
$ns(1) set Y_ 250
$ns(1) set Z_ 0

$ns(2) set X_ 400
$ns(2) set Y_ 250
$ns(2) set Z_ 0

$ns initial_node_pos $ns(0) 20
$ns initial_node_pos $ns(1) 20
$ns initial_node_pos $ns(2) 20

set udp [new Agent/UDP]
set null [new Agent/Null]

$ns attach-agent $ns(0) $udp
$ns attach-agent $ns(2) $null
$ns connect $udp $null

set cbr [new Application/Traffic/CBR]
$cbr set packetSize_ 512
$cbr set interval_ 0.1
$cbr attach-agent $udp
```

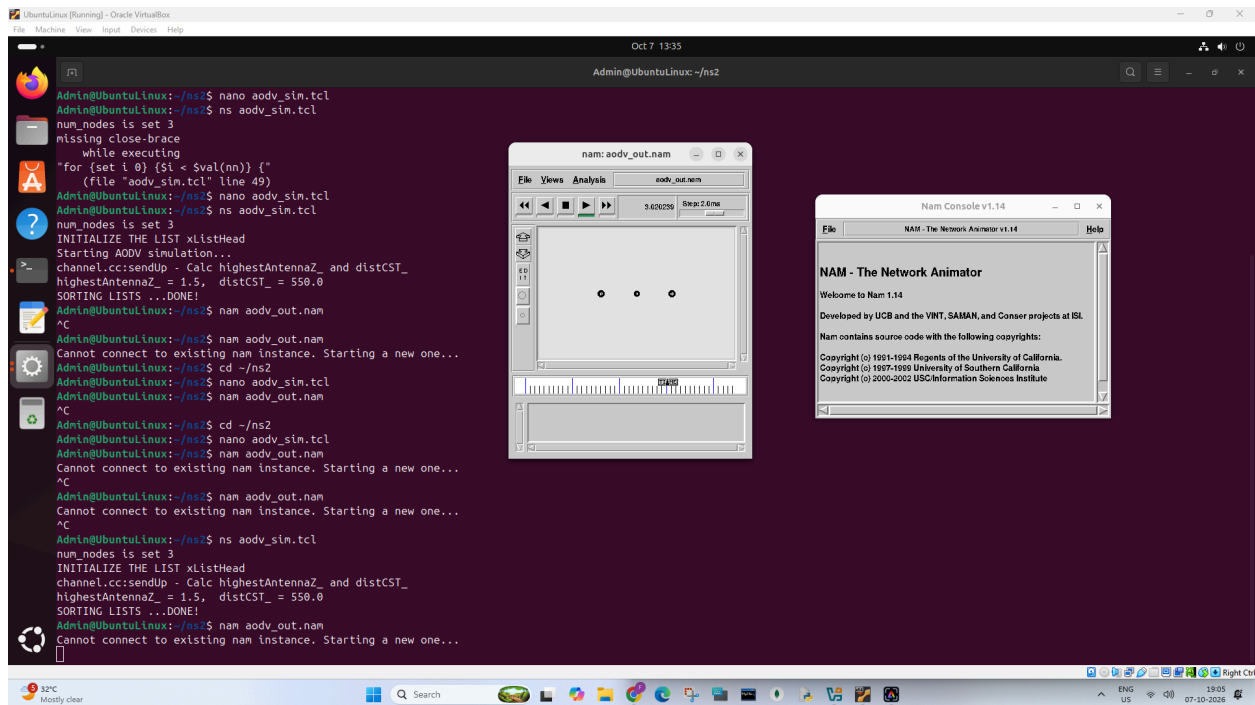
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Ns2 terminal commands & simulation:



The screenshot shows a terminal window titled 'UbuntuLinux (Running) - Oracle VM VirtualBox' with the prompt 'Admin@UbuntuLinux: ~/ns2'. The terminal displays the following commands and output:

```
Admin@UbuntuLinux:~/ns2$ nano aadv_sim.tcl
Admin@UbuntuLinux:~/ns2$ ns aadv_sim.tcl
num_nodes is set 3
missing close-brace
while executing
"for {set i 0} {$i < $val(nn)} {"
(file "aadv_sim.tcl" line 49)
Admin@UbuntuLinux:~/ns2$ nano aadv_sim.tcl
Admin@UbuntuLinux:~/ns2$ ns aadv_sim.tcl
num_nodes is set 3
INITIALIZE THE LIST xListHead
Starting AODV simulation...
channel.cc:sendUp - Calc highestAntennaZ_ and distCST_
highestAntennaZ_ = 1.5, distCST_ = 550.0
SORTING LISTS ...DONE!
Admin@UbuntuLinux:~/ns2$ nam aadv_out.nam
^C
Admin@UbuntuLinux:~/ns2$ nam aadv_out.nam
Cannot connect to existing nam instance. Starting a new one...
Admin@UbuntuLinux:~/ns2$ cd ~/ns2
Admin@UbuntuLinux:~/ns2$ nano aadv_sim.tcl
Admin@UbuntuLinux:~/ns2$ ns aadv_out.nam
Cannot connect to existing nam instance. Starting a new one...
^C
Admin@UbuntuLinux:~/ns2$ cd ~/ns2
Admin@UbuntuLinux:~/ns2$ nano aadv_sim.tcl
Admin@UbuntuLinux:~/ns2$ ns aadv_out.nam
Cannot connect to existing nam instance. Starting a new one...
^C
Admin@UbuntuLinux:~/ns2$ ns aadv_out.nam
Cannot connect to existing nam instance. Starting a new one...
^C
Admin@UbuntuLinux:~/ns2$ ns aadv_sim.tcl
num_nodes is set 3
INITIALIZE THE LIST xListHead
channel.cc:sendUp - Calc highestAntennaZ_ and distCST_
highestAntennaZ_ = 1.5, distCST_ = 550.0
SORTING LISTS ...DONE!
Admin@UbuntuLinux:~/ns2$ nam aadv_out.nam
Cannot connect to existing nam instance. Starting a new one...
```

Two windows are open over the terminal:

- nam: aadv_out.nam**: A window showing a network diagram with three nodes and a timeline at the bottom.
- Nam Console v1.14**: A window displaying the NAM - The Network Animator welcome message and copyright information.

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